

Math 104: Final Study Guide

How to Apply the Theorems of Real Analysis

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1 $\limsup$ and $\liminf$

Intuition

$\limsup s_n$ is the “eventual ceiling” of the sequence — the largest value that subsequences can cluster around. $\liminf s_n$ is the “eventual floor.” If they agree, the sequence converges to that value.

Think of it this way: $\limsup s_n = L$ means that (1) infinitely many terms get close to L , and (2) only finitely many terms exceed $L + \varepsilon$ for any $\varepsilon > 0$.

Two equivalent definitions:

1. **Subsequential limits:** $\limsup s_n = \sup\{\text{all subsequential limits of } \{s_n\}\}.$
2. **Tail sups:** $\limsup s_n = \inf_n \sup_{k \geq n} s_k = \lim_{n \rightarrow \infty} \sup_{k \geq n} s_k.$

The tail-sup formulation is often easier to compute: the sequence $a_n = \sup_{k \geq n} s_k$ is decreasing, so its limit always exists (possibly $\pm\infty$).

How to Apply

When you see $\limsup$ on an exam:

- To **compute** $\limsup s_n$: find all subsequential limits (cluster points), take the largest. Or compute $\sup_{k \geq n} s_k$ and take $n \rightarrow \infty$.
- To **show convergence**: prove $\limsup s_n = \liminf s_n$.
- To **bound a sequence**: $\limsup s_n \leq L$ means $s_n < L + \varepsilon$ eventually.
- In the **root/ratio tests**: the test computes $\limsup$ of $|a_n|^{1/n}$ or $|a_{n+1}/a_n|$ and compares it to 1.

Example

$s_n = (-1)^n(1 + 1/n)$. The subsequential limits are $\{-1, 1\}$. So $\limsup s_n = 1$ and $\liminf s_n = -1$.

Example

$s_n = \sin(n)$. The set of subsequential limits is $[-1, 1]$ (since $\{n \bmod 2\pi\}$ is dense in $[0, 2\pi]$). So $\limsup = 1$, $\liminf = -1$.

2 Metric Spaces and Topology

2.1 Open, Closed, Compact, Connected, Dense

Intuition

These five properties describe different “shapes” a subset can have:

- **Open:** every point has breathing room (an ε -ball fitting inside).
- **Closed:** the set “contains its boundary” — limits of sequences in the set stay in the set.
- **Compact:** the set is “finite-like” — you can always extract finite subcovers, convergent subsequences, etc. In $\mathbb{R}^n$: closed + bounded.
- **Connected:** the set is “one piece” — you can’t split it into two separated parts.
- **Dense:** the set is “everywhere” — every point of the ambient space is either in the set or arbitrarily close to it.

How to Apply

To show a set is open: For each $x \in E$, find $\varepsilon > 0$ with $B(x, \varepsilon) \subseteq E$. Alternatively, show E is a union of open balls, or show E^c is closed.

To show a set is closed: Show E contains all its limit points: if $x_n \in E$ and $x_n \rightarrow x$, then $x \in E$. Or show E^c is open.

To show a set is compact (in $\mathbb{R}^n$): Use Heine–Borel: show it’s closed AND bounded. For a general metric space, you must work from the definition (every open cover has a finite subcover).

To show a set is connected: In $\mathbb{R}$, show it’s an interval (no gaps). In general, assume $E = A \cup B$ with A, B separated and nonempty, derive a contradiction.

To show a set is dense: Show that for every $x \in X$ and every $\varepsilon > 0$, there exists a point of E within distance ε of x . Equivalently, show $\overline{E} = X$.

Common Pitfall

- A set can be **neither open nor closed** (e.g., $[0, 1)$ in $\mathbb{R}$) or **both open and closed** ($\emptyset$, $\mathbb{R}$, or any set in the discrete topology).
- **Bounded does not imply compact** in general metric spaces. Example: $(0, 1)$ is bounded but not compact. Heine–Borel only works in $\mathbb{R}^n$.
- **Closed and bounded $\neq$ compact** in arbitrary metric spaces. You need $\mathbb{R}^n$.
- An **infinite intersection of open sets** need not be open: $\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$.

2.2 Closure, Limit Points, and the Closure Formula

The closure $\overline{E} = E \cup E'$ where E' is the set of limit points. This is the smallest closed set containing E .

How to Apply

To find $\overline{E}$: Ask “what are the limit points of E ?” A point p is a limit point if every neighborhood of p contains a point of E different from p .

Example: $E = \{1/n : n \in \mathbb{N}\}$. The only limit point is 0 (every neighborhood of 0 contains some $1/n$). So $\overline{E} = E \cup \{0\} = \{0, 1, 1/2, 1/3, \dots\}$, which is closed.

2.3 Compactness: Why It Matters

Compactness is the most powerful tool in analysis. Whenever you have compactness, you get:

1. Every sequence has a convergent subsequence (in the set)
2. Every continuous function attains its max and min (Extreme Value Theorem)
3. Continuous functions are automatically uniformly continuous
4. Open covers have finite subcovers (the definition!)
5. The finite intersection property holds

How to Apply

The compactness trick: You need something to hold globally. Get a local version at each point x (using continuity, openness, etc.), which gives an open set U_x around x . These U_x cover the compact set K . Extract a **finite** subcover $U_{x_1}, \dots, U_{x_n}$. Now take $\delta = \min(\delta_{x_1}, \dots, \delta_{x_n})$, which works globally since the minimum of finitely many positive numbers is positive.

This is exactly how you prove “continuous on compact $\Rightarrow$ uniformly continuous.”

3 Cantor Set and Its Uses

The Cantor set C is the universal counterexample machine in real analysis.

Construction: Start with $[0, 1]$. Remove the middle third $(1/3, 2/3)$. From each remaining interval, remove its middle third. Repeat. $C = \bigcap_{n=0}^{\infty} C_n$.

Ternary characterization: $C = \{x \in [0, 1] : x \text{ has a base-3 expansion using only digits 0 and 2}\}$.

How to Apply

Use the Cantor set when you need:

- **An uncountable set with measure zero:** C is uncountable ($C \sim \{0, 1\}^{\mathbb{N}}$) but has

Lebesgue measure 0 (total length $(2/3)^n \rightarrow 0$).

- A **perfect set** (closed, no isolated points): $C' = C$.
- A **nowhere dense** closed set: C contains no interval, yet is uncountable.
- A **totally disconnected** compact set: the only connected subsets of C are singletons.
- Proof that **nonempty perfect sets are uncountable**: C is the canonical example.

4 Continuity and the Smoothness Hierarchy

4.1 The Three Faces of Continuity

Continuity has three equivalent definitions (for metric spaces):

1. ε - δ : $\forall \varepsilon > 0, \exists \delta > 0: d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$.
2. **Sequential**: $x_n \rightarrow x \Rightarrow f(x_n) \rightarrow f(x)$.
3. **Topological**: Preimages of open sets are open.

Use ε - δ for explicit proofs, sequential for contradictions (“if f weren’t continuous, there’d be a bad sequence”), and topological for clean abstract arguments.

4.2 Continuity vs. Uniform Continuity

Intuition

Continuous: at each point, you can find a δ that works for that point (different points may need different δ ’s).

Uniformly continuous: one δ works for ALL points simultaneously.

How to Apply

To show f is NOT uniformly continuous: Find two sequences x_n, y_n with $|x_n - y_n| \rightarrow 0$ but $|f(x_n) - f(y_n)| \not\rightarrow 0$.

Example: $f(x) = x^2$ on $\mathbb{R}$ is not uniformly continuous. Take $x_n = n$, $y_n = n + 1/n$. Then $|x_n - y_n| = 1/n \rightarrow 0$ but $|f(x_n) - f(y_n)| = |2 + 1/n^2| \rightarrow 2 \neq 0$.

Shortcut: If the domain is compact and f is continuous, you automatically get uniform continuity. So non-uniform-continuity can only happen on non-compact domains.

4.3 The Smoothness Hierarchy

$$C^\infty \subsetneq \cdots \subsetneq C^2 \subsetneq C^1 \subsetneq C^0$$

Each inclusion is strict. Know the separating examples:

- C^0 but not C^1 : $f(x) = |x|$ (continuous, not differentiable at 0).
- C^0 but not C^1 (subtler): $f(x) = x \sin(1/x)$, $f(0) = 0$.
- Differentiable but derivative not continuous: $f(x) = x^2 \sin(1/x)$, $f(0) = 0$. Here $f'(0) = 0$ exists but f' is not continuous at 0.
- C^∞ but not analytic: $f(x) = e^{-1/x^2}$ for $x \neq 0$, $f(0) = 0$.

5 Intermediate Value Theorem and Mean Value Theorem

5.1 IVT: Continuous Functions Hit Everything Between

Statement: If f is continuous on $[a, b]$ and c is between $f(a)$ and $f(b)$, then $f(p) = c$ for some $p \in (a, b)$.

How to Apply

Use IVT to show equations have solutions:

1. Define f so that your equation becomes $f(x) = 0$ (or $f(x) = c$).
2. Find a, b where $f(a) < 0$ and $f(b) > 0$ (or vice versa).
3. Conclude by IVT: $\exists p$ with $f(p) = 0$.

Example: Show $x^5 + x = 1$ has a solution. Let $f(x) = x^5 + x - 1$. $f(0) = -1 < 0$, $f(1) = 1 > 0$. By IVT, $\exists p \in (0, 1)$ with $f(p) = 0$.

Common Pitfall

IVT requires continuity on a **closed interval**. It fails for discontinuous functions (Dirichlet function takes values 0 and 1 but misses everything in between on any interval).

5.2 MVT: Derivatives Match Secant Lines

Statement: f continuous on $[a, b]$, differentiable on $(a, b) \Rightarrow \exists c \in (a, b)$ with $f'(c) = \frac{f(b) - f(a)}{b - a}$.

How to Apply

Use MVT to bound function values:

$$|f(b) - f(a)| = |f'(c)| \cdot |b - a| \leq \sup_{[a, b]} |f'| \cdot |b - a|.$$

This is the workhorse inequality of analysis. It says: “if the derivative is small, the function doesn’t change much.”

Use MVT to prove functions are constant: If $f' = 0$ on (a, b) , then for any $x, y \in (a, b)$: $|f(x) - f(y)| = |f'(c)||x - y| = 0$.

Use MVT to prove uniqueness of fixed points: If $f(x_1) = x_1$ and $f(x_2) = x_2$ with $x_1 \neq x_2$, MVT gives $f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = 1$. So if $f' \neq 1$ everywhere, at most one fixed point.

5.3 Generalized MVT and L'Hôpital's Rule

The Generalized MVT replaces the denominator $b - a$ with $g(b) - g(a)$:

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

L'Hôpital's rule follows from GMVT: if $f(x), g(x) \rightarrow 0$ as $x \rightarrow a$, then

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)} \rightarrow \frac{\lim f'}{\lim g'}.$$

How to Apply

L'Hôpital recipe:

1. Verify you have $0/0$ or ∞/∞ form.
2. Differentiate numerator and denominator separately.
3. If the new limit exists, it equals the original.
4. If you get $0/0$ again, apply L'Hôpital again.

5.4 Taylor's Theorem

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n)}(c)}{n!} (x-a)^n$$

The last term is the **Lagrange remainder** — it tells you how good the polynomial approximation is.

How to Apply

To bound the error of a Taylor approximation: The error is $|R_n| = \frac{|f^{(n)}(c)|}{n!} |x-a|^n$ for some unknown c . Bound $|f^{(n)}(c)| \leq M$ on the interval, giving $|R_n| \leq \frac{M}{n!} |x-a|^n$.

6 Series

6.1 Choosing a Convergence Test

How to Apply

Decision tree for $\sum a_n$:

1. Does $a_n \rightarrow 0$? If not, the series **diverges** (necessary condition).
2. Does it look like a known series (p -series, geometric)? Use **comparison**.
3. Are there factorials or exponentials? Try the **ratio test**.
4. Is $a_n = f(n)$ for some nice f ? Try the **root test** (always at least as good as ratio).
5. Is it decreasing and positive? Try **Cauchy condensation**.
6. Is it alternating? Try the **alternating series test** (Leibniz).

6.2 Absolute vs. Conditional Convergence

Absolutely convergent: $\sum |a_n| < \infty$. This is the “safe” type — you can rearrange terms, group them, multiply series, etc.

Conditionally convergent: $\sum a_n$ converges but $\sum |a_n| = \infty$. This is fragile — the **Riemann rearrangement theorem** says you can rearrange the terms to make the series converge to *any* value you want (or diverge).

How to Apply

To determine which type:

1. First check if $\sum |a_n|$ converges (using comparison, root, ratio tests).
2. If yes: absolutely convergent.
3. If $\sum |a_n|$ diverges but $\sum a_n$ converges (often via alternating series test): conditionally convergent.

Example

$\sum (-1)^n/n$: converges by alternating series test, but $\sum 1/n$ diverges (harmonic). So it converges **conditionally**.

$\sum (-1)^n/n^2$: $\sum 1/n^2$ converges (p -series, $p = 2 > 1$). So it converges **absolutely**.

6.3 Radius of Convergence

For a power series $\sum c_n(x-a)^n$, the radius of convergence is:

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}}$$

How to Apply

To find R :

- **Root test method:** Compute $\limsup |c_n|^{1/n}$, take reciprocal.
- **Ratio test shortcut:** If $\lim |c_{n+1}/c_n|$ exists, then $R = \lim |c_n/c_{n+1}|$.
- **Always check endpoints** $x = a \pm R$ separately — convergence there is not determined by R .

Example

$\sum x^n/n$: $c_n = 1/n$, so $|c_n|^{1/n} = n^{-1/n} \rightarrow 1$, giving $R = 1$. At $x = 1$: $\sum 1/n$ diverges. At $x = -1$: $\sum (-1)^n/n$ converges (alternating). So: converges on $[-1, 1)$.

7 Sequences and Series of Functions

7.1 Pointwise vs. Uniform Convergence

Intuition

Pointwise: At each individual point x , $f_n(x) \rightarrow f(x)$. Different points may converge at very different rates.

Uniform: ALL points converge at the same rate. After some N , the entire graph of f_n is within ε of the graph of f .

How to Apply

To show uniform convergence: Compute $\|f_n - f\|_\infty = \sup_{x \in E} |f_n(x) - f(x)|$ and show it $\rightarrow 0$.

Common technique: find the maximum of $|f_n(x) - f(x)|$ by taking the derivative and setting it to zero.

To show convergence is NOT uniform:

1. Find the supremum $\sup_x |f_n(x) - f(x)|$ and show it does NOT go to 0.
2. Or: the limit function is discontinuous but each f_n is continuous (uniform limit of continuous functions must be continuous).

Example

$f_n(x) = x^n$ on $[0, 1]$. Pointwise limit: $f(x) = 0$ for $x \in [0, 1)$, $f(1) = 1$. The limit is discontinuous, but each f_n is continuous. So the convergence is **not uniform**.

Alternatively: $\sup_{x \in [0, 1]} |x^n - f(x)| \geq |f_n(1 - 1/n) - 0| = (1 - 1/n)^n \rightarrow 1/e \neq 0$.

7.2 Weierstrass M-Test

Statement: If $|f_n(x)| \leq M_n$ for all x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly.

How to Apply

To apply the M-test:

1. For each n , find $M_n = \sup_{x \in E} |f_n(x)|$.
2. Check if $\sum M_n$ converges.
3. If yes, $\sum f_n$ converges uniformly on E .

Example: $\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$ on $\mathbb{R}$. We have $|\sin(nx)/n^2| \leq 1/n^2 = M_n$. Since $\sum 1/n^2$ converges, the series converges uniformly by M-test.

Common Pitfall

The M-test is a **sufficient** condition, not necessary. The converse is false: a series can converge uniformly even if the M-test fails. But the M-test is the go-to first attempt.

8 Uniform Convergence and Interchange of Limits

The central theme: **when can we swap the order of two limit operations?** The answer is almost always: when convergence is uniform.

8.1 Uniform Convergence + Continuity

Theorem: Uniform limit of continuous functions is continuous.

$$f_n \text{ continuous, } f_n \rightarrow f \text{ uniformly} \implies f \text{ continuous}$$

How to Apply

Use this to prove a function is continuous: Write it as the uniform limit of continuous functions (often partial sums of a series). Use the M-test to verify uniform convergence.

Use the contrapositive to show convergence is not uniform: If $f = \lim f_n$ is discontinuous but each f_n is continuous, the convergence cannot be uniform. (This is the x^n argument.)

8.2 Uniform Convergence + Integration

Theorem: If $f_n \rightarrow f$ uniformly and each f_n is integrable, then f is integrable and

$$\int_a^b f \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n \, dx.$$

Intuition

Uniform convergence lets you “pass the limit inside the integral.” This is the easiest interchange theorem — it just works, no extra hypotheses needed (beyond uniform convergence and integrability of each f_n).

8.3 Uniform Convergence + Differentiation

Theorem: If f_n are differentiable, f'_n converges uniformly to g , and $f_n(x_0)$ converges for some x_0 , then $f_n \rightarrow f$ uniformly and $f' = g$.

Common Pitfall

This is the trickiest interchange. You need:

1. Uniform convergence of the **derivatives** f'_n (not the functions!).
2. Pointwise convergence of f_n at **at least one point**.

Uniform convergence of f_n alone is NOT enough for $(\lim f_n)' = \lim f'_n$. Example: $f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \rightarrow 0$ uniformly, but $f'_n(x) = \sqrt{n} \cos(nx)$ diverges.

8.4 Summary Table

Operation	Hypothesis	Conclusion
Continuity	f_n cts, $f_n \rightarrow f$ unif.	f is cts
Integration	$f_n \in \mathcal{R}$, $f_n \rightarrow f$ unif.	$\int f = \lim \int f_n$
Differentiation	$f'_n \rightarrow g$ unif., $f_n(x_0)$ conv.	$f' = g = \lim f'_n$

9 Equicontinuity and Arzelà–Ascoli

9.1 What is Equicontinuity?

Intuition

A family $\{f_n\}$ is equicontinuous if ALL functions in the family are “uniformly continuous with the same δ .” In other words, the modulus of continuity is uniform across the entire family, not just across points.

Single function, uniform continuity: One δ works for all **points**.

Family, equicontinuity: One δ works for all **points** AND all **functions in the family**.

How to Apply

To verify equicontinuity: Find a single δ (depending only on ε) such that $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$ for ALL n .

Common sufficient conditions:

- All f_n are Lipschitz with the **same** Lipschitz constant L : $\delta = \varepsilon/L$.
- All f_n are differentiable with $|f'_n| \leq M$ (uniformly bounded derivatives): $\delta = \varepsilon/M$ by MVT.
- $f_n \rightarrow f$ uniformly and each f_n is continuous $\Rightarrow$ equicontinuous.

9.2 Arzelà–Ascoli: Compactness for Function Spaces

Statement: If K is compact and $\{f_n\} \subset \mathcal{C}(K)$ is **pointwise bounded** and **equicontinuous**, then $\{f_n\}$ has a **uniformly convergent** subsequence.

Intuition

Arzelà–Ascoli is the Bolzano–Weierstrass theorem for functions. In $\mathbb{R}^n$, bounded sequences have convergent subsequences. In $\mathcal{C}(K)$, pointwise bounded + equicontinuous families have convergent subsequences.

Think of it as: “bounded + equicontinuous = compact in function space.”

How to Apply

To apply Arzelà–Ascoli: Check three things:

1. **Domain is compact:** K must be compact.
2. **Pointwise bounded:** $\forall x \in K, \exists M_x: |f_n(x)| \leq M_x$ for all n .
3. **Equicontinuous:** One δ works for all n and all points.

Then conclude: there exists a uniformly convergent subsequence.

Example

Show that $\{f_n\}$ where $f_n : [0, 1] \rightarrow \mathbb{R}$, $|f_n| \leq 10$, and $|f'_n| \leq 5$ for all n has a uniformly convergent subsequence.

Solution: Domain $[0, 1]$ is compact. Pointwise bounded: $|f_n(x)| \leq 10$. Equicontinuous: $|f_n(x) - f_n(y)| \leq 5|x - y|$ by MVT, so $\delta = \varepsilon/5$ works for all n . By Arzelà–Ascoli, done.

10 Riemann Integrability

10.1 The Big Picture

A bounded function f on $[a, b]$ is Riemann integrable if and only if it can be “squeezed” between step functions that are arbitrarily close together.

10.2 The Criterion

$f \in \mathcal{R}[a, b] \Leftrightarrow \forall \varepsilon > 0, \exists \text{ partition } P \text{ with } U(P, f) - L(P, f) < \varepsilon.$

How to Apply

To show $f \in \mathcal{R}$: Construct a partition that makes $U - L$ small. Strategy:

1. Identify where f has big oscillation (“bad” intervals).
2. Make the total length of bad intervals very small.
3. On the “good” intervals, f doesn’t vary much, so $M_i - m_i$ is small.
4. Total: $U - L \leq (\text{bad contribution}) + (\text{good contribution}).$
5. Bad: $\leq 2\|f\|_\infty \cdot (\text{total bad length}).$ Good: $\leq \varepsilon \cdot (b - a).$

To show $f \notin \mathcal{R}$: Show $U(P, f) - L(P, f) \geq c > 0$ for ALL partitions P .

10.3 Sufficient Conditions (know these!)

1. f continuous on $[a, b] \Rightarrow f \in \mathcal{R}$. (Uniform continuity controls oscillation.)
2. f monotone on $[a, b] \Rightarrow f \in \mathcal{R}$. (Oscillation on $[x_{i-1}, x_i]$ is $f(x_i) - f(x_{i-1})$, and these telescope.)
3. f bounded with discontinuities forming a **zero set** $\Rightarrow f \in \mathcal{R}$.

A set E is a **zero set** (measure zero) if for every $\varepsilon > 0$, E can be covered by countably many intervals with total length $< \varepsilon$. Examples: finite sets, countable sets (like $\mathbb{Q}$), the Cantor set.

10.4 The Three Key Counterexamples

1. **Thomae’s function** $T(x)$: $T(p/q) = 1/q$ (in lowest terms), $T(\text{irrational}) = 0$.
 - Continuous at every irrational (given ε , only finitely many rationals p/q with $1/q \geq \varepsilon$ in any bounded interval).
 - Discontinuous at every rational.
 - $\in \mathcal{R}$ because the discontinuity set $\mathbb{Q}$ is a zero set.
 - Shows: you can be discontinuous on a dense set and still be integrable.

2. **Dirichlet function** $D(x) = \mathbf{1}_{\mathbb{Q}}(x)$:

- Discontinuous everywhere.
- $U(P, D) = 1$ and $L(P, D) = 0$ for every partition (every interval contains both rationals and irrationals).
- $\notin \mathcal{R}$.
- Shows: the “everywhere discontinuous” case fails.

3. $1/\sqrt{x}$ on $(0, 1]$:

- Unbounded near 0.
- Not Riemann integrable (Riemann integration requires bounded functions on closed intervals).
- But the improper integral $\int_0^1 x^{-1/2} dx = 2$ exists.
- Shows: unboundedness breaks Riemann integrability, but improper integrals can rescue the situation.

How to Apply

Geometric interpretation of $U(P, f) - L(P, f) < \varepsilon$: Draw the function. The upper sum is the area of rectangles sitting on top of the curve; the lower sum sits below. Their difference is the total area of the “cracks” between upper and lower rectangles. Integrability means these cracks can be made arbitrarily thin.

11 Putting It All Together: Common Exam Problem Types

1. **“Show f is uniformly continuous”:** If the domain is compact, just show continuity. Otherwise, find δ independent of the point.
2. **“Show $\sum f_n$ converges uniformly”:** Try the Weierstrass M-test first.
3. **“Show the limit is continuous/integrable/differentiable”:** Show uniform convergence (or for differentiation: uniform convergence of derivatives), then apply the interchange theorem.
4. **“Does the family have a convergent subsequence?”:** Check Arzelà–Ascoli conditions (compact domain, pointwise bounded, equicontinuous).
5. **“Show f is Riemann integrable”:** Check if f is continuous, monotone, or has a zero set of discontinuities.
6. **“Find the radius of convergence”:** Use $R = 1/\limsup |c_n|^{1/n}$ or the ratio test. Then check endpoints.
7. **“Show a sequence converges”:** Show $\limsup = \liminf$, or show it’s Cauchy, or show it’s monotone and bounded.
8. **“Show a set is compact/connected/dense”:** For compact in $\mathbb{R}^n$: closed and bounded. For connected in $\mathbb{R}$: show it’s an interval. For dense: show $\overline{E} = X$.